IAT New Test 9

Question paper, answer key
and worked solutions

> **MANUAL REVIEW DRAFT**
> 2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
> The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Revised 2026-11-01 · IAT27_TEST_09 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

A fern bears one type of spore, and a free-living prothallus develops from it. Which conclusion follows?

A) It is homosporous; its prothallus is the gametophyte

B) It is heterosporous; the prothallus is sporophyte

C) Its spores are diploid gametes

D) Its prothallus makes spores by meiosis

Q2

+4 / -1

A floral formula records K(5) C(5) A5 G̲(2), with the underline denoting a superior ovary. Which interpretation is consistent?

A) Free sepals and two free carpels

B) Five fused sepals, five fused petals and a bicarpellary syncarpous superior ovary

C) An inferior ovary and ten stamens

D) A monocarpellary ovary with free petals

Q3

+4 / -1

The enzyme-rate figure compares an uninhibited curve with the curve under inhibitor X. Which inhibition mechanism best fits both Km and Vmax?

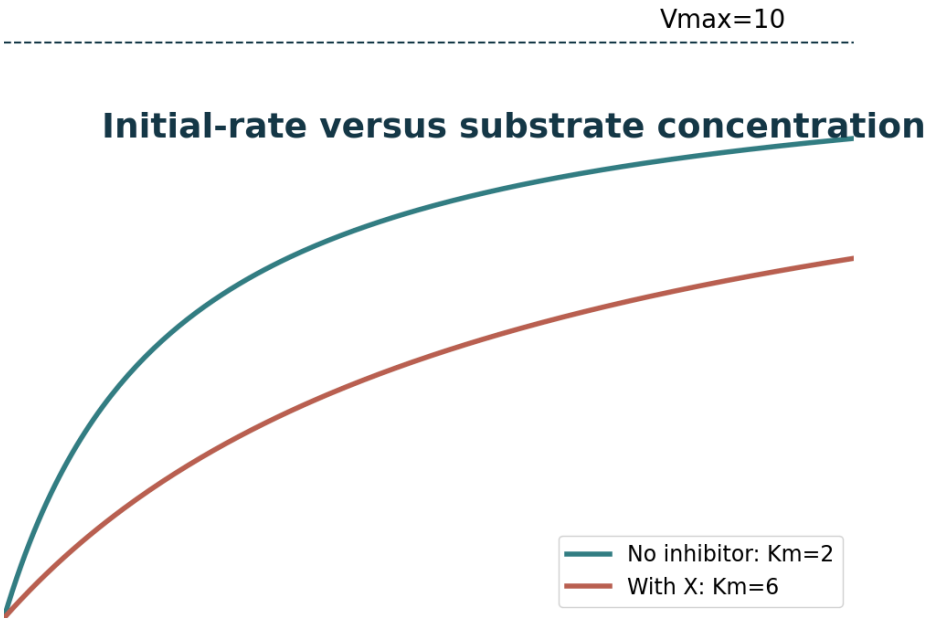


Figure for Q3

A) Irreversible loss of enzyme

B) Uncompetitive inhibition lowering both parameters

C) Competitive inhibition that can be overcome by high substrate

D) Activation of catalytic turnover

Q4

+4 / -1

In the leaf cross-section, bundle-sheath cells form a chloroplast-rich ring around each vascular bundle and mesophyll surrounds the ring. Which conclusion best fits?

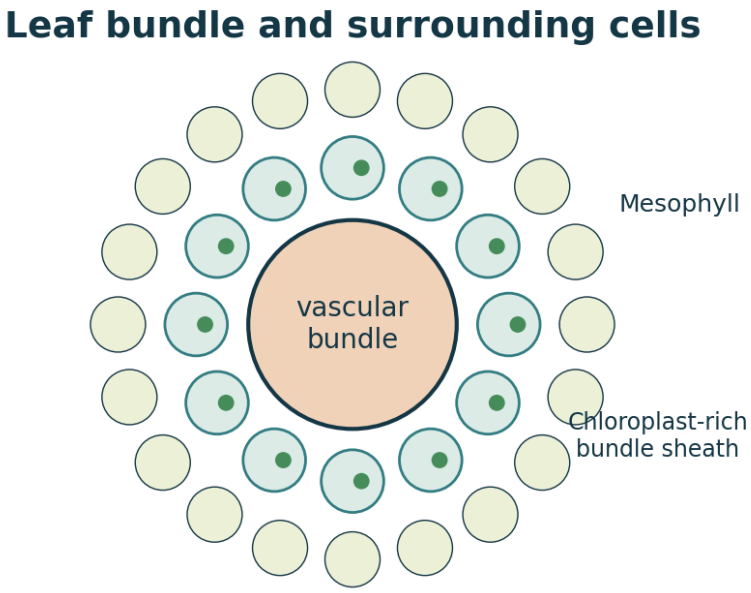


Figure for Q4

A) CAM storage is certain

B) Only C3 fixation can occur

C) The bundle sheath lacks chloroplasts

D) Kranz anatomy supports spatial C4 carbon concentration

Q5

+4 / -1

A postsynaptic neuron responds to externally applied transmitter but not to a presynaptic action potential. Its axon still conducts when directly stimulated. What is the most localised defect?

A) Transmitter release at the presynaptic terminal

B) Loss of all postsynaptic receptors

C) Failure of axonal Na^+ channels throughout the postsynaptic neuron

D) Inability of the transmitter to bind the intact receptor

Q6

+4 / -1

After fertilisation in an angiosperm, what do an ovule and an ovary generally become, respectively?

A) Embryo and endosperm

B) Seed and fruit

C) Fruit and seed

D) Seed coat and pollen grain

Q7

+4 / -1

An X-linked recessive disorder affects a father and not his partner, who is homozygous normal. What is expected among their children?

A) Every son affected

B) Half of sons affected

C) Every daughter a carrier and every son unaffected

D) Every daughter affected

Q8

+4 / -1

A cell culture loses p53 function. Following DNA damage it continues through G1/S and accumulates mutations. What best explains its greater cancer risk?

A) It automatically gains a new chromosome each division

B) Repair becomes unnecessary

C) Mitosis becomes meiosis

D) The damage checkpoint fails to pause or eliminate compromised cells

Q9

+4 / -1

A plasmid has ampicillin resistance and a disrupted tetracycline-resistance gene when insert is present. Transformants first grow on ampicillin. What screen selects likely recombinants?

A) Replica plate and retain colonies that fail to grow on tetracycline

B) Keep only colonies growing on both drugs

C) Keep cells that never grew on ampicillin

D) Discard all ampicillin-resistant colonies

Q10

+4 / -1

In an energy-flow measurement, producers fix 10,000 kJ; 2,000 kJ enters herbivores, which lose 1,200 kJ as heat and pass 300 kJ to predators. How much herbivore energy enters detritus?

A) 300 kJ

B) 500 kJ

C) 800 kJ

D) 1,500 kJ

Q11

+4 / -1

A replication fork advances right. The upper template runs 3′→5′ toward the fork, and its new daughter extends continuously toward the fork. Which statement is correct?

A) New DNA grows 3′→5′

B) Both daughters extend continuously in the fork direction

C) The upper daughter is leading and grows 5′→3′

D) The lower daughter must be synthesized 3′→5′

Q12

+4 / -1

In the lac-operon table, strain X expresses lacZ without lactose but shuts off in glucose; strain Y responds normally to both conditions. X has a repressor that cannot bind operator but intact glucose control. What explains X?

A) A defective promoter preventing all transcription

B) An enzyme unable to cleave lactose

C) A permanently inactive polymerase

D) Constitutive operator-level derepression with catabolite repression retained

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Q13

+4 / -1

A time-series measurement shows that a rise in circulating thyroid hormone is followed by a fall in TSH while external conditions are unchanged. Which feedback link is supported?

A) Thyroid hormone negatively feeds back on pituitary TSH output

B) TSH suppresses thyroid hormone irreversibly

C) Thyroid hormone is made by the pituitary

D) The thyroid axis has only positive feedback

Q14

+4 / -1

A cell-free translation mix has intact mRNA, ribosomes and charged tRNAs. A drug blocks movement of the ribosome by one codon after peptide-bond formation. Which stage directly fails?

A) Initiation of transcription

B) Elongation translocation

C) Aminoacyl-tRNA synthetase charging

D) DNA ligation

Q15

+4 / -1

For the same habitat type, species richness follows S=cA^0.25. If sampled area rises by a factor of 16, by what factor does expected richness rise?

A) 4

B) 8

C) 16

D) 2

Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

The H–O–H angle is about 104.5°, below the tetrahedral ideal. Which electron-pair explanation is best?

A) No lone pair on oxygen

B) Three bonding pairs on oxygen

C) Two lone pairs exert stronger repulsion than bonding pairs and compress the angle

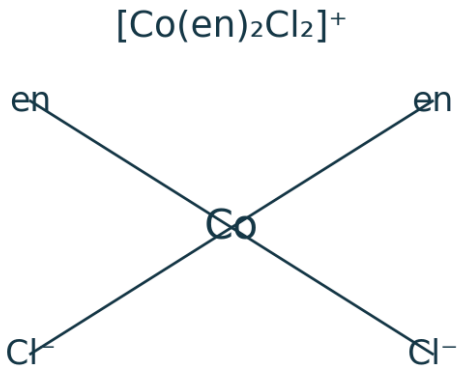
D) The molecule is linear but rotates rapidly

Q17

+4 / -1

Use the coordination sketch and its donor-site labels to find the metal oxidation state and coordination number.

Coordination sphere and donor sites



Each en makes two donor bonds; each Cl⁻ makes one

Figure for Q17

A) +2 and 4

B) +2 and 6

C) +3 and 4

D) +3 and 6

Q18

+4 / -1

Which ion is d^0 and hence lacks d–d transitions within its own d subshell?

A) Sc^{3+}

B) Ti^{3+}

C) V^{3+}

D) Cr^{3+}

Q19

+4 / -1

A hydrogen atom emits a photon in the 3→2 transition. If $E_n = -13.6/n^2$ eV and $hc = 1240$ eV·nm, its wavelength is closest to?

A) 486 nm

B) 656 nm

C) 434 nm

D) 122 nm

Q20

+4 / -1

Why can the first ionisation energy of oxygen be lower than that of nitrogen despite oxygen being to its right in period 2?

A) Oxygen has a completely empty 2p shell

B) Nitrogen has a paired 2p orbital

C) Oxygen has a paired 2p electron; nitrogen has a more stable half-filled $2p^3$ arrangement

D) Nitrogen has a larger principal quantum number

Q21

+4 / -1

A+B→C is 1:1. Starting with 0.10 mol A and 0.16 mol B, C is isolated at 80% yield. All isolated C then reacts by 2C→D at 50% yield. How many moles of D are isolated?

A) 0.020

B) 0.040

C) 0.050

D) 0.080

Q22

+4 / -1

Compare p-nitrobenzoic acid and p-methoxybenzoic acid in water. Which is generally more acidic, and why?

A) Methoxy acid: +M donation stabilises its anion

B) Nitro acid: electron withdrawal stabilises its conjugate base

C) Equal: para groups cannot affect resonance

D) Methoxy acid: inductive donation alone dominates

Q23

+4 / -1

Which single compound can undergo self-aldol addition under basic conditions?

A) Benzaldehyde

B) Formaldehyde

C) Propanal

D) Benzophenone

Q24

+4 / -1

Which compound has the strongest acidity among the listed phenols?

A) p-Methoxyphenol

B) Phenol

C) p-Methylphenol

D) p-Nitrophenol

Q25

+4 / -1

An unknown amine reacts with benzenesulfonyl chloride in alkali to give a sulfonamide insoluble in alkali; it does not dissolve after acidification. Which class is most consistent?

A) Secondary amine

B) Primary amine

C) Tertiary amine

D) Ammonium salt without an amine

Q26

+4 / -1

At a pH far above its isoelectric point, a simple amino acid without an ionisable side chain predominantly has what net charge?

A) +2

B) −1

C) 0

D) +1

Q27

+4 / -1

Using the pure-component and observed pressures shown in the figure at its marked liquid composition, what deviation occurs?

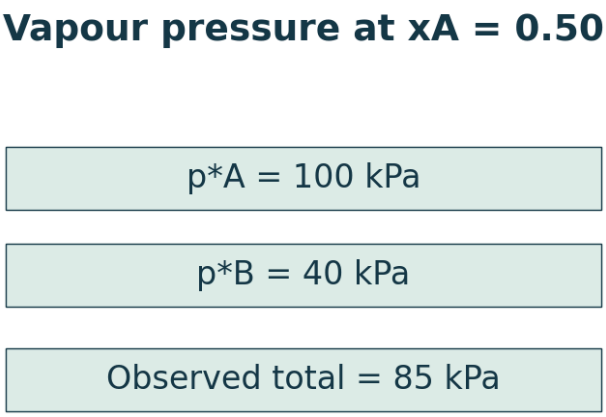


Figure for Q27

A) Negative, 15 kPa

B) Zero, ideal 85 kPa

C) Positive, 15 kPa

D) Positive, 30 kPa

Q28

+4 / -1

A concentration cell $Cu|Cu^{2+}(0.010\ M)||Cu^{2+}(1.0\ M)|Cu$ operates at 298 K. With E° identical and 0.0591 V in the Nernst factor, what is Ecell?

A) 0.0296 V

B) 0.0591 V

C) 0.1182 V

D) 0 V

Q29

+4 / -1

For a reaction with Ea=50 kJ mol^{-1}, what is approximately k(310 K)/k(300 K)? Take R=8.314 J mol^{-1} K^{-1}.

A) 1.91

B) 1.09

C) 1.48

D) 4.8

Q30

+4 / -1

For $N_2+3H_2\rightleftharpoons 2NH_3$ at fixed T, the initial quotient Q is less than K. What is the immediate spontaneous direction?

A) Reverse, because ammonia is a product

B) No change, because T is fixed

C) Both directions equally

D) Forward until Q approaches K

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

Let A be invertible 3×3 with det(A)=3. Form B by adding twice row 1 to row 2, then multiplying row 3 by −2. What is det(adj B)?

A) −36

B) −6

C) 36

D) 6

Q32

+4 / -1

On a square grid, how many right/up paths go from (0,0) to (3,3) without passing through (1,1)?

A) 12

B) 10

C) 6

D) 8

Q33

+4 / -1

For real x, which inputs make $\ln(x^2-2)$ defined if this expression is obtained by f(g(x)), with $f(t)=\ln(t-1)$ and $g(x)=x^2-1$?

A) $(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$

B) $[-\sqrt{2},\sqrt{2}]$

C) All nonzero x

D) $(-\infty,-1)\cup(1,\infty)$

Q34

+4 / -1

In the Argand plane, what locus satisfies $|z-(1+i)|=|z-(3-i)|$?

A) x+y=2

B) x−y=2

C) x+y=1

D) x−y=1

Q35

+4 / -1

Evaluate ∫₀^π x sin x dx.

A) π/2

B) 2π

C) π

D) 0

Q36

+4 / -1

Project the point P=(1,2) orthogonally onto the oblique line 3x+4y−10=0 in the plane. Which ordered pair is obtained?

A) (4/5,9/5)

B) (0,5/2)

C) (1,7/4)

D) (22/25,46/25)

Q37

+4 / -1

For f(x)=ax+b if x<1, and x² if x≥1, which (a,b) make f differentiable at x=1?

A) (2,−1)

B) (1,0)

C) (2,1)

D) (−2,3)

Q38

+4 / -1

Two groups each have 10 values with means 2 and 4 and population variances 1 and 1. What is the variance of all 20 values?

A) 1

B) 2

C) 3

D) 4

Q39

+4 / -1

Find a such that $\lim_{x\to 0} (\sin(ax)-2x)/x^3$ exists as a finite real number.

A) 0

B) 1

C) 2

D) 3

Q40

+4 / -1

Urn U_1 is chosen with probability 1/3 and has 2 red, 1 blue; U_2 with probability 2/3 has 1 red, 3 blue. After observing red, what is $P(U_1|\text{red})$?

A) 1/3

B) 2/3

C) 1/2

D) 4/7

Q41

+4 / -1

If $\tan\theta=1/2$ with θ acute, find $\tan(2\theta)$.

A) 4/3

B) 3/4

C) 1

D) 2

Q42

+4 / -1

For $x>0$ and $y>0$, solve $dy/dx=(x^2+y^2)/(xy)$ with $y(1)=1$. What is $y(e)$?

A) $e\sqrt{2}$

B) $e\sqrt{3}$

C) 2e

D) 3e

Q43

+4 / -1

For which real k are both roots of $x^2-2kx+(k+3)=0$ real and in the open interval (0,3)?

A) $k \geq (1+\sqrt{13})/2$

B) $(1+\sqrt{13})/2 < k \leq 12/5$

C) $(1+\sqrt{13})/2 \leq k < 12/5$

D) $k > 12/5$

Q44

+4 / -1

Every line in the family $(1+t)x+(2-t)y=3+t$ passes through which point?

A) (1,1)

B) (0,3/2)

C) (2,0)

D) (5/3,2/3)

Q45

+4 / -1

For which k are (1,2), (3,6), and (k,10) collinear?

A) 4

B) 5

C) 6

D) 7

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

The figure gives the static-friction threshold for a block on a horizontal floor. While the block remains at rest under a horizontal 4 N applied force, what is its friction?

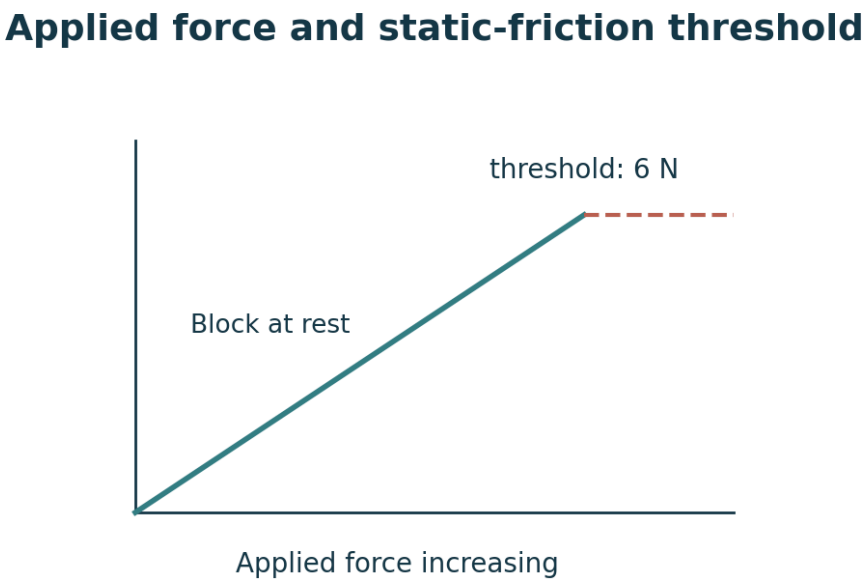


Figure for Q46

A) 4 N opposite the applied force

B) 6 N opposite the force

C) 2 N opposite the force

D) Zero until sliding

Q47

+4 / -1

A one-dimensional conservative force is $F(x)=x^2-4x$ newtons, with x measured in metres. What is its signed work as a particle moves from $x=0$ to $x=4$ m?

A) $+32/3$ J

B) $-32/3$ J

C) 0

D) -16 J

Q48

+4 / -1

At a planet surface r=R, escape speed is v. What minimum launch speed reaches r=4R with zero speed there, ignoring atmosphere and rotation?

A) v/2

B) v/4

C) √3v/2

D) 3v/4

Q49

+4 / -1

A stretched rod has longitudinal strain +0.002 and Poisson ratio 0.30. What is the transverse strain?

A) +0.006

B) −0.006

C) +0.0006

D) −0.0006

Q50

+4 / -1

Two identical hot bodies at 80°C and 60°C cool in air at 20°C under the same Newton cooling constant. What is the ratio of their initial cooling-rate magnitudes, hotter to cooler?

A) 3:2

B) 2:1

C) 4:3

D) 1:1

Q51

+4 / -1

The kinetic-theory sketch shows N molecules of mass m in a cube of side L, each with mean-square x-velocity component $\langle c_x^2 \rangle$. What pressure acts on the wall normal to x?

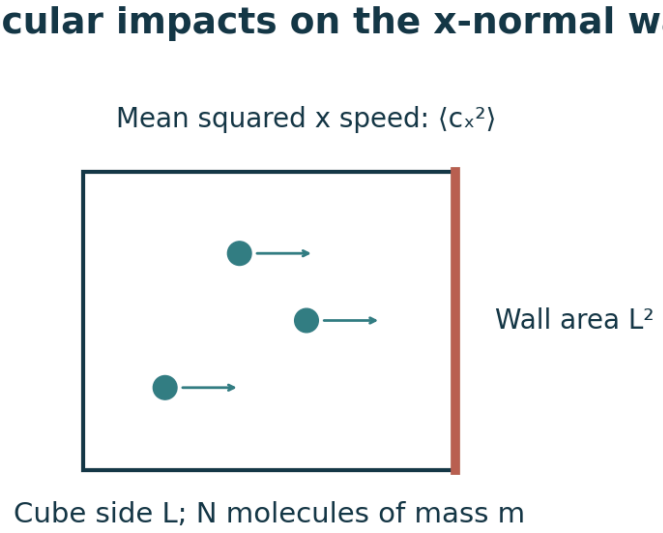


Figure for Q51

A) $Nm\langle c_x^2 \rangle / L^2$

B) $Nm\langle c_x^2 \rangle / L^3$

C) $Nm\langle c_x^2 \rangle / (3L^3)$

D) $Nm\langle c_x^2 \rangle / (6L^3)$

Q52

+4 / -1

For SHM with total mechanical energy E and amplitude A, what is kinetic energy at displacement x=A/2?

A) E/4

B) E/2

C) 3E/4

D) E

Q53

+4 / -1

The before/after diagram shows a charged capacitor isolated from the battery and then filled with the indicated dielectric. What are its new voltage and energy relative to V_0, U_0?

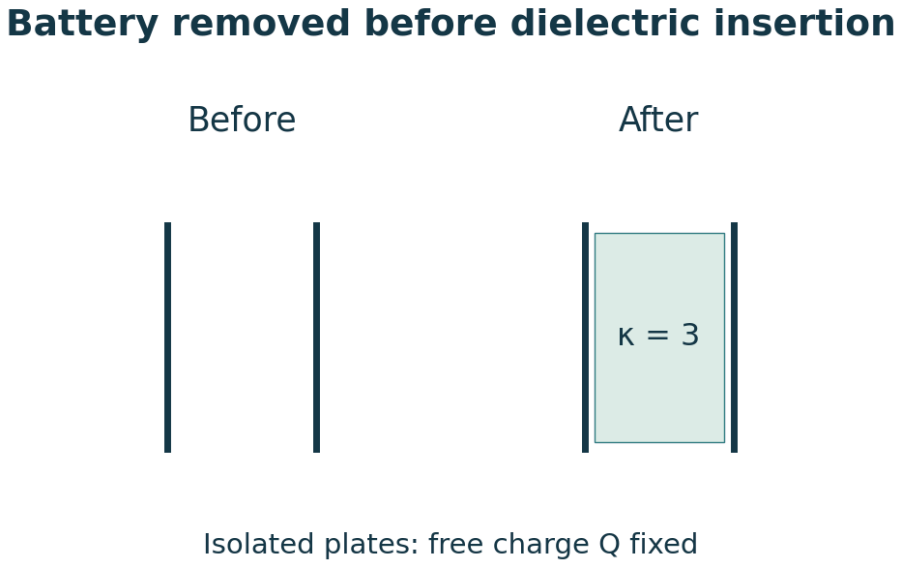


Figure for Q53

A) $3V_0, 3U_0$

B) V_0, U_0

C) $V_0/3, 3U_0$

D) $V_0/3, U_0/3$

Q54

+4 / -1

A positive charge moves with velocity $v\,\hat{x}$ through $B(\hat{y}+\hat{z})$. Which uniform electric field makes its total Lorentz force zero?

A) $vB(\hat{y}-\hat{z})$

B) $vB(\hat{z}-\hat{y})$

C) $vB(\hat{y}+\hat{z})$

D) Zero

Q55

+4 / -1

A closed surface encloses charges +2q, −q and +3q. A fourth charge −5q is outside. What is its net electric flux?

A) 0

B) $4q/\varepsilon_0$

C) $-q/\varepsilon_0$

D) $-5q/\varepsilon_0$

Q56

+4 / -1

A resistor network has two identical 6 Ω resistors in parallel between A and B, then a 2 Ω resistor in series from B to C. What is equivalent resistance between A and C?

A) 3 Ω

B) 4 Ω

C) 5 Ω

D) 8 Ω

Q57

+4 / -1

An object 30 cm left of a converging lens (f=10 cm) first forms an image; a plane mirror 20 cm right of the lens reflects those rays back through the lens. Where is the final image after the second lens passage?

A) 5 cm left of lens

B) 10 cm left of lens

C) 15 cm left of lens

D) 50/3 cm left of lens

Q58

+4 / -1

At fixed frequency above threshold, light intensity incident on a photocathode doubles while collector voltage is high enough to saturate. What happens to saturation photocurrent and stopping potential?

A) Current doubles; stopping potential unchanged

B) Both double

C) Current unchanged; potential doubles

D) Both unchanged

Q59

+4 / -1

The hydrogen-level figure marks n=4,3,2,1. Which downward transition has the longest emitted wavelength among 4→3, 4→2, 3→2, 2→1?

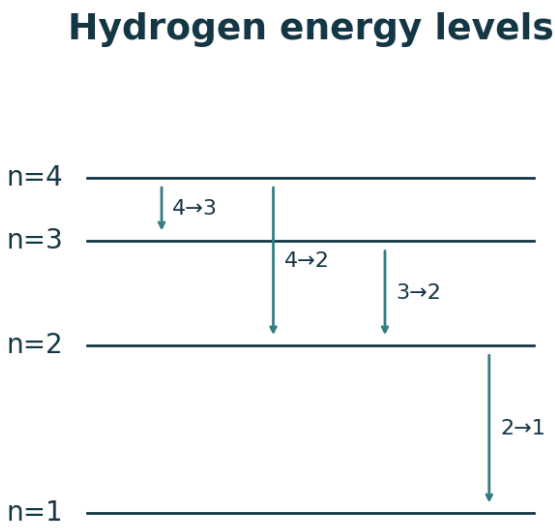


Figure for Q59

A) 4→2

B) 3→2

C) 4→3

D) 2→1

Q60

+4 / -1

Two nearby coils have corresponding dotted ends. The left coil current toward its marked dot decreases. What is the initial induced current direction at the marked dot of the right coil?

A) Out of the right dot

B) Zero because coils do not touch

C) Alternates randomly

D) Into the right dot

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**A**
2	**B**
3	**C**
4	**D**
5	**A**
6	**B**
7	**C**
8	**D**
9	**A**
10	**B**
11	**C**
12	**D**
13	**A**
14	**B**
15	**D**

Chemistry

Q	Answer
16	**C**
17	**D**
18	**A**
19	**B**
20	**C**
21	**A**
22	**B**
23	**C**
24	**D**
25	**A**
26	**B**
27	**C**
28	**B**
29	**A**
30	**D**

Mathematics

Q	Answer
31	**C**
32	**D**
33	**A**
34	**B**
35	**C**
36	**D**
37	**A**
38	**B**
39	**C**
40	**D**
41	**A**
42	**B**
43	**C**
44	**D**
45	**B**

Physics

Q	Answer
46	**A**
47	**B**
48	**C**
49	**D**
50	**A**
51	**B**
52	**C**
53	**D**
54	**A**
55	**B**
56	**C**
57	**D**
58	**A**
59	**C**
60	**D**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 42 medium / 9 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · medium

Diversity In the Living World / pteridophyte spore type

Answer: (A) It is homosporous; its prothallus is the gametophyte.

Why: A single spore class defines homospory. Germination gives the haploid prothallus bearing gametangia; the spore-bearing leafy fern is the sporophyte. This identifies the free-living stage from its origin and reproductive structures rather than from its size.

Why not the others: (B) invents two spore classes and mislabels the generation; (C) spores are products of meiosis, not gametes; (D) spore meiosis belongs to the sporophyte sporangium.

Key point: Track ploidy and generation separately.

Q2

Biology · easy

Structural Organisation In Plants And Animals / floral formula inference

Answer: (B) Five fused sepals, five fused petals and a bicarpellary syncarpous superior ovary.

Why: Parentheses indicate fusion within a whorl: K(5), C(5), and G(2) are fused groups. A5 means five stamens; the stated convention marks the ovary superior.

Why not the others: (A) ignores the parentheses; (C) changes both ovary position and stamen count; (D) contradicts petal and carpel fusion.

Key point: Read the symbols and position convention independently.

Q3

Biology · medium

Cell: Structure And Functions / enzyme inhibition

Answer: (C) Competitive inhibition that can be overcome by high substrate.

Why: The same asymptotic Vmax means sufficiently abundant substrate restores the maximal rate, while a larger substrate concentration is needed for half-maximal rate. Competition for the active site has this pattern.

Why not the others: (A) usually lowers functional enzyme and Vmax; (B) changes both Km and Vmax downward; (D) would increase maximal output rather than preserve it.

Key point: Compare both Km and Vmax, not a single rate.

Q4

Biology · medium

Plant Physiology / C3 versus C4 anatomy

Answer: (D) Kranz anatomy supports spatial C4 carbon concentration.

Why: The pictured concentric organisation is Kranz anatomy. Initial fixation in mesophyll can be followed by CO_2 release near RuBisCO in bundle sheath, reducing oxygenation under suitable conditions.

Why not the others: (A) CAM relies on temporal rather than this spatial separation; (B) conflicts with the usual C4 interpretation; (C) contradicts the drawn chloroplasts.

Key point: Anatomy supports a pathway inference, not a measured rate.

Q5

Biology · hard

Human Physiology / neural synapse and receptor

Answer: (A) Transmitter release at the presynaptic terminal.

Why: Direct transmitter application verifies that postsynaptic receptor and downstream excitability can work. Direct axon stimulation verifies conduction. The missing response only when presynaptic input is used points to release or delivery from that terminal.

Why not the others: (B) and (D) conflict with the direct application response; (C) conflicts with direct axonal conduction.

Key point: Use bypass experiments to locate a failed step.

Q6

Biology · medium

Reproduction / seed and fruit development

Answer: (B) Seed and fruit.

Why: The fertilised ovule matures as a seed containing the embryo; the enclosing ovary generally develops into the fruit. Integuments contribute the seed coat, but that is only one part of the seed.

Why not the others: (A) names contents rather than the enclosing structures; (C) reverses the two; (D) substitutes a substructure and an unrelated male gametophyte.

Key point: Follow the whole structures through maturation.

Q7

Biology · easy

Genetics And Evolution / sex-linked transmission

Answer: (C) Every daughter a carrier and every son unaffected.

Why: The father gives his Y to sons and affected X to daughters. The mother supplies a normal X to each child; daughters are heterozygous carriers and sons receive their only X from her.

Why not the others: (A) and (B) incorrectly send the paternal X to sons; (D) overlooks the normal maternal X.

Key point: Separate paternal transmission by child sex.

Q8

Biology · medium

Biology In Human Welfare / cancer and cell-cycle link

Answer: (D) The damage checkpoint fails to pause or eliminate compromised cells.

Why: p53 supports arrest and downstream repair or apoptosis responses after damage. Without that restraint, damaged genomes enter replication and pass variants to descendants, increasing opportunity for malignant selection.

Why not the others: (A) is not automatic; (B) repair remains needed; (C) changes the division type without support.

Key point: A failed checkpoint permits propagation of damage.

Q9

Biology · medium

Biotechnology / recombinant selection control

Answer: (A) Replica plate and retain colonies that fail to grow on tetracycline.

Why: Ampicillin growth selects plasmid-containing cells. A correctly inserted fragment disables tetracycline resistance, so ampicillin-positive but tetracycline-negative colonies are candidate recombinants; verify insert by a separate assay.

Why not the others: (B) favours intact vector; (C) lacks evidence of plasmid uptake; (D) discards the selected transformants.

Key point: Selection and insertional screening answer different questions.

Q10

Biology · hard

Ecology / food-web energy accounting

Answer: (B) 500 kJ.

Given: Input 2000 kJ and two known outputs.

Formula/Principle: Energy balance: detritus = input − heat − predation.

Step 1: 2000−1200−300=500 kJ.

Why: Conservation across the herbivore box gives 2,000 kJ in minus 1,200 kJ heat minus 300 kJ predator transfer, leaving 500 kJ for detritus. This bookkeeping does not assume an arbitrary 10% rule.

Why not the others: (A) repeats predator transfer; (C) subtracts only heat; (D) subtracts only detrital or trophic routes incorrectly.

Key point: Account for every output of the same compartment.

Q11

Biology · medium

Genetics And Evolution / molecular replication direction

Answer: (C) The upper daughter is leading and grows 5′→3′.

Why: DNA polymerase extends the 3′ end, giving synthesis 5′→3′. Reading the upper template 3′→5′ toward the advancing fork allows continuous leading-strand synthesis in that direction; the other template requires fragments.

Why not the others: (A) reverses polymerase direction; (B) ignores antiparallel templates; (D) reverses synthesis direction on the lagging strand.

Key point: The 5′→3′ synthesis rule applies to both daughters.

Q12

Biology · easy

Genetics And Evolution / transcription regulation

Answer: (D) Constitutive operator-level derepression with catabolite repression retained.

Why: A repressor that cannot occupy operator cannot shut the operon off in the absence of inducer. The intact glucose control still lowers activation under glucose, explaining both observed features.

Why not the others: (A) and (C) predict no expression; (B) concerns substrate processing rather than operator occupancy and does not explain constitutive transcription.

Key point: Two regulatory layers can act independently.

Q13

Biology · medium

Human Physiology / endocrine feedback

Answer: (A) Thyroid hormone negatively feeds back on pituitary TSH output.

Why: An increase in the downstream hormone followed by a reduced upstream drive under otherwise stable conditions is the expected sign of negative feedback. The graph alone does not identify every molecular intermediate.

Why not the others: (B) reverses the observed causal direction; (C) misplaces the hormone source; (D) predicts TSH increasing rather than falling.

Key point: Use direction and sequence of change to identify feedback sign.

Q14

Biology · medium

Cell: Structure And Functions / ribosome translation context

Answer: (B) Elongation translocation.

Why: Peptide bond formation can occur, so the immediate blocked step is translocation during elongation, when mRNA/tRNAs shift relative to the ribosome to expose the next codon.

Why not the others: (A) is upstream and mRNA is already present; (C) is excluded by charged tRNAs and timing; (D) is unrelated to peptide synthesis.

Key point: Map the inhibited movement to its translation stage.

Q15

Biology · medium

Ecology / species-area relationship

Answer: (D) 2.

Given: $S=cA^{0.25}$ and $A_2/A_1=16$.

Formula/Principle: $S_2/S_1=(A_2/A_1)^{0.25}$.

Step 1: $16^{(1/4)}=2$.

Why: The species-area law uses a sublinear exponent, so the richness ratio is $16^{0.25}=2$, not the area ratio. c cancels for two areas in the same comparison.

Why not the others: (B) treats the exponent as one half; (C) uses three quarters; (D) assumes direct proportionality.

Key point: Take the ratio before substituting the area factor.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · medium

Chemical Bonding and Molecular Structure / VSEPR angle with lone pair

Answer: (C) Two lone pairs exert stronger repulsion than bonding pairs and compress the angle.

Why: Oxygen has two O–H bond pairs and two lone pairs in approximately tetrahedral electron-pair geometry. Lone-pair repulsions push bonding pairs closer, reducing the observed bond angle.

Why not the others: (A) omits two lone pairs; (B) counts bonds incorrectly; (D) contradicts bent molecular geometry.

Key point: Distinguish electron-pair geometry from molecular shape.

Q17

Chemistry · hard

Coordination compounds / coordination number and charge

Answer: (D) +3 and 6.

Why: Let Co oxidation state be x. Two neutral en and two Cl^- give $x-2=+1$, so $x=+3$. Each en donates two sites (four total) and two chlorides one each, giving coordination number six.

Why not the others: (A) and (B) miscount charge; (C) counts ligand molecules rather than donor atoms.

Key point: Charge and donor count require separate calculations.

Q18

Chemistry · medium

The d- & f-block elements / colour and d-electron count

Answer: (A) Sc^{3+}.

Why: Neutral Sc is $[Ar]3d^14s^2$; removal of three electrons gives $[Ar]3d^0$. With no d electron, a d–d excitation is unavailable, though other colour mechanisms can exist in compounds.

Why not the others: (B) Ti^{3+} is d^1; (C) V^{3+} is d^2; (D) Cr^{3+} is d^3.

Key point: Compute the ionic d count before invoking colour.

Q19

Chemistry · easy

Structure of the Atom / hydrogen transition wavelength

Answer: (B) 656 nm.

Given: Hydrogen levels n=3 and 2.

Formula/Principle: $\Delta E = 13.6(1/2^2 - 1/3^2)$ eV; $\lambda = hc/\Delta E$.

Step 1: $\Delta E = 1.889$ eV.

Why: The energy gap is $13.6(1/4 - 1/9) = 1.889$ eV, so $\lambda = 1240/1.889 \approx 656$ nm. The emitted photon energy is the positive difference of level energies. Since wavelength varies inversely with energy, choosing a larger transition gap would give a shorter line.

Why not the others: (A) and (C) correspond to different Balmer transitions; (D) is much shorter wavelength and requires a larger gap.

Key point: Use energy difference, then convert by hc/E.

Q20

Chemistry · medium

Classification of Elements and Periodicity in Properties / ionisation anomaly

Answer: (C) Oxygen has a paired 2p electron; nitrogen has a more stable half-filled $2p^3$ arrangement.

Why: Nitrogen $2p^3$ puts one electron in each p orbital; oxygen $2p^4$ introduces a pair. Repulsion in that pair eases removal relative to the half-filled nitrogen configuration, overcoming the simple across-period trend locally.

Why not the others: (A) reverses configuration; (B) gives the pair to nitrogen; (D) both occupy n=2.

Key point: Local subshell stability modifies periodic trends.

Q21

Chemistry · medium

Some Basic Concepts of Chemistry / limiting reagent in stages

Answer: (A) 0.020.

Given: A=0.10 mol, B=0.16 mol; yields 80%, then 50%.

Formula/Principle: $n(C) = 0.80 \min(nA, nB)$; $n(D) = 0.50\, n(C)/2$.

Step 1: $n(C) = 0.080$ mol.

Why: A limits the first step at 0.10 mol theoretical C; 80% isolation gives 0.080 mol C. The second stoichiometry gives 0.040 mol theoretical D and 50% of this is 0.020 mol. Apply each yield to its own stage.

Why not the others: (B) omits second yield; (C) applies yield incorrectly before 2:1 conversion; (D) reports isolated C as D.

Key point: Carry isolated intermediate to the next reaction.

Q22

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / resonance versus inductive acidity

Answer: (B) Nitro acid: electron withdrawal stabilises its conjugate base.

Why: The para nitro group withdraws electron density through resonance and induction, stabilising the negatively charged benzoate relative to the acid. Para methoxy donates through resonance and tends to oppose this stabilisation.

Why not the others: (A) calls electron donation stabilising for an anion; (C) ignores conjugation through the ring; (D) reverses the key para resonance effect.

Key point: Judge acidity by relative conjugate-base stability.

Q23

Chemistry · medium

Aldehydes, ketones, and carboxylic acids / aldol eligibility

Answer: (C) Propanal.

Why: Propanal has an alpha carbon bearing hydrogens, so enolate formation and attack on another propanal molecule are possible. The listed aromatic or one-carbon carbonyl compounds lack alpha hydrogen in the needed position.

Why not the others: (A) benzaldehyde lacks an alpha carbon; (B) formaldehyde lacks one; (D) benzophenone has phenyl groups directly attached to carbonyl carbon, without alpha hydrogen.

Key point: Check for at least one alpha hydrogen on the enolate donor.

Q24

Chemistry · hard

Alcohols, phenols, and ethers / phenol acidity

Answer: (D) p-Nitrophenol.

Why: A para nitro substituent withdraws electron density and stabilises phenoxide by resonance and induction. Methoxy and methyl are electron donating in the relevant comparison, while phenol lacks substituent stabilisation.

Why not the others: (A) methoxy resonance donation destabilises phenoxide; (B) lacks nitro stabilisation; (C) methyl donation does not stabilise the anion.

Key point: Compare resonance stabilisation of the conjugate base.

Q25

Chemistry · medium

Organic compounds containing nitrogen / amine-class test

Answer: (A) Secondary amine.

Why: A secondary amine forms a sulfonamide with no N–H remaining, so it does not form the soluble anion in alkali. A primary sulfonamide retains an acidic N–H and dissolves in alkali; a tertiary amine lacks the N–H needed to form this sulfonamide.

Why not the others: (B) product should dissolve in base; (C) cannot furnish the described sulfonamide; (D) does not explain the reaction product.

Key point: Hinsberg solubility depends on residual sulfonamide N–H.

Q26

Chemistry · easy

Biomolecules / amino acid charge

Answer: (B) −1.

Why: High pH deprotonates the carboxyl to COO^- and the ammonium group to neutral NH_2, giving a net −1. This describes the predominant form sufficiently above both relevant pKa values.

Why not the others: (A) overcounts positive charge; (C) is the zwitterion near its isoelectric region; (D) is favoured at strongly acidic pH.

Key point: Compare pH with protonation equilibria of both groups.

Q27

Chemistry · medium

Solutions / Raoult deviation

Answer: (C) Positive, 15 kPa.

Given: xA=xB=0.50; p*A=100, p*B=40, observed=85 kPa.

Formula/Principle: pideal=xAp*A+xBp*B.

Step 1: pideal=70 kPa.

Why: Raoult ideal pressure at equimolar composition is 0.5(100)+0.5(40)=70 kPa. Observed pressure exceeds ideal by 15 kPa, the signature of positive deviation from Raoult’s law.

Why not the others: (A) reverses sign; (B) treats observed as ideal; (D) computes the magnitude incorrectly.

Key point: Compare observed pressure to weighted pure-component pressures.

Q28

Chemistry · medium

Electrochemistry / concentration-cell potential

Answer: (B) 0.0591 V.

Given: Cu^{2+} concentrations differ by factor 100.

Formula/Principle: $E=(0.0591/n)\log(\text{cathode/anode})$, $n=2$.

Step 1: $E=0.02955\times2=0.0591$ V.

Why: The higher-concentration Cu^{2+} half-cell is reduced, the lower is oxidised. $E=(0.0591/2)\log(1/0.010)=0.0591$ V; standard potentials cancel. The net cell reaction transfers copper ions from the concentrated side toward the dilute side, reducing the concentration difference.

Why not the others: (A) misses log ratio 100; (C) omits electron count two; (D) overlooks concentration difference.

Key point: Use concentration ratio and n=2.

Q29

Chemistry · hard

Chemical Kinetics / Arrhenius temperature comparison

Answer: (A) 1.91.

Given: $T_1=300$ K, $T_2=310$ K, $Ea=50000$ J mol^{-1}.

Formula/Principle: $\ln(k_2/k_1)=Ea/R(1/T_1-1/T_2)$.

Step 1: Exponent ≈0.647.

Why: The two-temperature Arrhenius ratio has $\ln(k_2/k_1)=(Ea/R)(1/T_1-1/T_2)\approx0.647$, hence $\exp(0.647)\approx1.91$. The factor is dimensionless and greater than one as temperature rises. Converting activation energy to joules is necessary because the stated gas constant uses joules.

Why not the others: (B) underestimates exponential sensitivity; (C) uses a smaller exponent; (D) exceeds the calculated factor.

Key point: Work with reciprocal absolute temperatures.

Q30

Chemistry · medium

Equilibrium / reaction quotient shift

Answer: (D) Forward until Q approaches K.

Why: $Q<K$ means the present products-to-reactants activity ratio is below its equilibrium value. Forward reaction increases product relative to reactants until the equilibrium condition is approached.

Why not the others: (A) uses product identity instead of Q; (B) fixed T fixes K but not Q; (C) equilibrium is the condition $Q=K$, not $Q<K$.

Key point: Compare reaction quotient with equilibrium constant.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · hard

Matrices & determinants / determinant row transform

Answer: (C) 36.

Given: A is 3×3, det A=3.

Formula/Principle: det B=−2 det A; det(adj B)=(det B)^(3−1).

Step 1: det B=−6.

Worked derivation:

$$\det(\operatorname{adj} B) = (\det B)^2 = (-2 \cdot 3)^2 = 36.$$

Why: The row addition preserves determinant and the scaling makes det B=−6. For any invertible 3×3 matrix, det(adj B)=(det B)², giving 36; the sign disappears only at the second operation.

Why not the others: (A) retains the determinant sign after squaring; (B) stops at det B; (D) takes the magnitude of det B but omits the adjugate identity.

Key point: Track row effects, then apply the dimension-dependent adjugate identity.

Q32

Mathematics · medium

Permutations and Combinations / path count avoiding a point

Answer: (D) 8.

Given: Three R and three U steps; forbidden point (1,1).

Formula/Principle: Nallowed=C(6,3)−C(2,1)C(4,2).

Step 1: 20−2·6=8.

Worked derivation:

$$N = \binom{6}{3} - \binom{2}{1}\binom{4}{2} = 20 - 12 = 8.$$

Why: There are 6 choose 3 =20 paths overall. A path through (1,1) has 2 choose 1 ways to reach it and 4 choose 2 ways afterward, 12 forbidden paths. Subtract to get 8; giving the selected 8 valid paths.

Why not the others: (A) counts forbidden paths; (B) miscounts the complement; (C) understates valid paths.

Key point: Use product counting through the forbidden point.

Q33

Mathematics · easy

Relations & functions / composition domain

Answer: (A) $(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$.

Given: $f(g(x))=\ln(x^2-2)$.

Formula/Principle: Logarithm argument $x^2-2>0$.

Step 1: $x<-\sqrt{2}$ or $x>\sqrt{2}$.

Worked derivation:

$$x^2-2>0 \iff x\in(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty).$$

Why: Composition gives $\ln(x^2-2)$, whose argument must be strictly positive. Solving $x^2>2$ gives two open rays; equality is excluded because $\ln 0$ is undefined.

Why not the others: (B) reverses the inequality and includes endpoints; (C) permits arguments below zero; (D) uses the domain of g rather than the composed logarithm.

Key point: Apply the outer logarithm restriction after composition.

Q34

Mathematics · medium

Complex Numbers / modulus locus

Answer: (B) $x-y=2$.

Given: Points (1,1) and (3,−1).

Formula/Principle: $(x-1)^2+(y-1)^2=(x-3)^2+(y+1)^2$.

Step 1: $4x-4y-8=0$.

Worked derivation:

$$(x-1)^2+(y-1)^2=(x-3)^2+(y+1)^2 \iff x-y=2.$$

Why: For $z=x+iy$, equate squared distances to (1,1) and (3,−1). Simplification gives $-4x+4y+8=0$, hence $x-y=2$. The locus is the perpendicular bisector of the two points.

Why not the others: (A) has wrong orientation; (C) changes the constant; (D) is a shifted parallel line.

Key point: Equidistance becomes a linear equation after squares cancel.

Q35

Mathematics · medium

Integration / symmetry of definite integral

Answer: (C) π.

Given: Integral from 0 to π.

Formula/Principle: ∫x sin x dx=−x cos x+sin x.

Step 1: Upper endpoint π; lower endpoint 0.

Worked derivation:

$$\int_0^{\pi} x \sin x \, dx = [-x \cos x + \sin x]_0^{\pi} = \pi.$$

Why: Integration by parts with u=x and dv=sin x dx gives −x cos x+sin x. At π this is π, at zero it is zero.

Why not the others: (A) halves the boundary term; (B) doubles it; (D) overlooks the nonzero endpoint contribution.

Key point: Keep the boundary term in integration by parts.

Q36

Mathematics · medium

Three-dimensional geometry / foot of perpendicular

Answer: (D) (22/25,46/25).

Given: P=(1,2), normal=(3,4), signed offset=1.

Formula/Principle: H=P−[(3·1+4·2−10)/25](3,4).

Step 1: H=(22/25,46/25).

Worked derivation:

$$H = (1,2) - \frac{1}{25}(3,4) = \left(\frac{22}{25}, \frac{46}{25}\right).$$

Why: The signed line expression at P is 3+8−10=1. Subtract 1/(3²+4²) times the normal vector (3,4), yielding (22/25,46/25), the selected foot.

Why not the others: (A) is off the projection; (B) lies elsewhere on the line; (C) is not a normal displacement.

Key point: Project along the oblique line normal.

Q37

Mathematics · medium

Differentiation / parameterised differentiability

Answer: (A) (2,−1).

Given: Piecewise join at x=1.

Formula/Principle: Continuity a+b=1; derivative match a=2.

Step 1: b=−1.

Worked derivation:

$$a+b=1, \quad a=2 \quad \Longrightarrow \quad (a,b)=(2,-1).$$

Why: Continuity first requires a+b=1. Equality of one-sided derivatives requires a=2, so b=−1. Both conditions are necessary; matching only value is insufficient.

Why not the others: (B) is continuous but slopes differ; (C) has matching derivative but a+b≠1; (D) is continuous but left slope −2.

Key point: Check value and slope at the joining point.

Q38

Mathematics · easy

Statistics / combined-group variance

Answer: (B) 2.

Given: Equal group sizes, means 2 and 4, variances 1 each.

Formula/Principle: V=average[Vgroup+(μgroup−μ)²].

Step 1: μ=3; V=(1+1+1+1)/2=2.

Worked derivation:

$$\mu=3, \qquad V=\tfrac{1}{2}[1+(2-3)^2+1+(4-3)^2]=2.$$

Why: The combined mean is 3. Within-group variance contributes 1; the two group means differ from 3 by ±1, adding between-group variance 1. Total population variance is 2.

Why not the others: (A) ignores between-group spread; (C) adds a mean gap without squaring/weighting correctly; (D) treats the full mean gap as a deviation.

Key point: Combine within and between variance.

Q39

Mathematics · medium

Limit & continuity / continuity parameter

Answer: (C) 2.

Given: Limit of [sin(ax)−2x]/x³.

Formula/Principle: sin(ax)=ax−a³x³/6+O(x⁵).

Step 1: Finite iff a−2=0.

Worked derivation:

$$\frac{\sin(ax)-2x}{x^3}=\frac{a-2}{x^2}-\frac{a^3}{6}+O(x^2);\quad a=2.$$

Why: Expand sin(ax)=ax−a³x³/6+higher terms. The coefficient of x must vanish for the quotient to avoid x^{-2} divergence, so a=2. The resulting finite limit is −8/6.

Why not the others: (A), (B) and (D) leave a nonzero (a−2)x term, producing divergence upon division by x³.

Key point: Cancel lower-order terms before taking a higher-order limit.

Q40

Mathematics · medium

Probability / Bayes with unequal priors

Answer: (D) 4/7.

Given: P(U₁)=1/3, P(R|U₁)=2/3; P(U₂)=2/3, P(R|U₂)=1/4.

Formula/Principle: Bayes P(U₁|R)=P(U₁)P(R|U₁)/P(R).

Step 1: P(R)=2/9+1/6=7/18.

Worked derivation:

$$P(U_1\mid R)=\frac{(1/3)(2/3)}{(1/3)(2/3)+(2/3)(1/4)}=\frac{4}{7}.$$

Why: The red joint probabilities are (1/3)(2/3)=2/9 and (2/3)(1/4)=1/6. Their sum is 7/18, so posterior U₁ is (2/9)/(7/18)=4/7. The observation favours U₁ despite its smaller prior because red is substantially more likely there.

Why not the others: (A) repeats the prior; (B) uses U₁ likelihood alone; (C) ignores unequal priors and urn compositions.

Key point: Normalise joint probabilities after the observation.

Q41

Mathematics · medium

Trigonometric functions / compound-angle identity

Answer: (A) 4/3.

Given: tan θ=1/2.

Formula/Principle: tan 2θ=2tanθ/(1−tan²θ).

Step 1: 1/(1−1/4)=4/3.

Worked derivation:

$$\tan 2\theta = \frac{2(1/2)}{1-(1/2)^2} = \frac{4}{3}.$$

Why: Apply tan 2θ=2t/(1−t²) with t=1/2: numerator 1, denominator 3/4, so 4/3. The acute condition fixes a consistent quadrant but the algebra already determines the value.

Why not the others: (B) inverts the result; (C) ignores the denominator; (D) doubles t incorrectly.

Key point: Use the full double-angle denominator.

Q42

Mathematics · medium

Differential equations / homogeneous substitution

Answer: (B) e√3.

Given: For x,y>0, y′=x/y+y/x.

Formula/Principle: Set y=xv; then v+xv′=1/v+v.

Step 1: v dv=dx/x, so v²=2ln x+C; v(1)=1 fixes C=1.

Worked derivation:

$$y = x\sqrt{1+2\ln x}, \qquad y(e) = e\sqrt{3}.$$

Why: This homogeneous differential equation becomes separable after putting y=xv. The initial point selects the positive square-root branch, which matters when evaluating y at e.

Why not the others: (A) omits the initial-value contribution; (C) uses an incorrect constant; (D) squares the required root.

Key point: Substitute y=xv, integrate, then apply the positive branch.

Q43

Mathematics · hard

Quadratic Equations / discriminant and parameter

Answer: (C) (1+√13)/2≤k<12/5.

Given: Polynomial has sum 2k, product k+3, vertex x=k.

Formula/Principle: Δ=4(k²−k−3)≥0; positive branch k≥(1+√13)/2.

Step 1: At x=3, p(3)=12−5k>0 and k<3.

Step 2: Hence lower endpoint included, upper excluded.

Worked derivation:

$$k \in \left[\frac{1+\sqrt{13}}{2}, \frac{12}{5}\right).$$

Why: Discriminant nonnegative and positive roots force k≥(1+√13)/2. To keep both roots below 3, the vertex k must lie below 3 and the polynomial at x=3 must be positive: 12−5k>0, so k<12/5. At the lower end the repeated root lies inside (0,3), while the upper end places a root at excluded x=3.

Why not the others: (A) includes arbitrarily large k; (B) wrongly excludes the valid repeated root and includes the endpoint 3; (D) makes the polynomial at 3 negative.

Key point: Combine discriminant, vertex position, endpoint signs and strict boundaries.

Q44

Mathematics · medium

Coordinate geometry / line family intersection

Answer: (D) (5/3,2/3).

Given: Line family holds for every t.

Formula/Principle: x+2y−3=0 and x−y−1=0.

Step 1: Solve simultaneous equations.

Worked derivation:

$$x = y+1, \quad y+1+2y = 3 \implies (x,y) = (5/3, 2/3).$$

Why: Collect constant and t terms: x+2y−3+t(x−y−1)=0. A common point satisfies both x+2y=3 and x−y=1; solving gives x=5/3,y=2/3, the common point in option D.

Why not the others: (A) fails the t coefficient; (B) fails the second constraint; (C) fails the constant constraint.

Key point: Set both polynomial coefficients in t to zero.

Q45

Mathematics · medium

Vectors / collinearity test

Answer: (B) 5.

Given: First two points define y=2x.

Formula/Principle: For y=10, x=10/2.

Step 1: k=5.

Worked derivation:

$$\frac{6-2}{3-1}=2, \quad y-2=2(x-1) \implies 10=2k \implies k=5.$$

Why: The first two points have slope (6−2)/(3−1)=2. The line through (1,2) is y=2x; setting y=10 gives x=5. This is option B.

Why not the others: (A), (C) and (D) do not satisfy the line equation y=2x at y=10.

Key point: Use one line equation and substitute the third point.

Physics

Method, reasoning and common errors

Q46

Physics · medium

Laws of Motion / limiting friction and impending slip

Answer: (A) 4 N opposite the applied force.

Given: Applied 4 N < limiting 6 N.

Formula/Principle: Horizontal equilibrium: f_s=Fapplied while at rest.

Step 1: f_s=4 N, opposite the force.

Worked derivation:

$$\sum F_x = 0 \implies f_s = F = 4\,\text{N} < 6\,\text{N}.$$

Why: For a stationary block static friction adjusts to match the applied horizontal force up to its limiting value. At 4 N, below the measured 6 N limit, force balance requires 4 N opposite the applied force.

Why not the others: (B) confuses the maximum with actual friction; (C) leaves unbalanced force; (D) would let the block accelerate before the observed threshold.

Key point: Static friction is self-adjusting below its limit.

Q47

Physics · medium

Work, Energy and Power / position-dependent work

Answer: (B) −32/3 J.

Given: $F=x^2-4x$, $0 \le x \le 4$.

Formula/Principle: $W=\int_0^4 F(x)\,dx$.

Step 1: $W=[x^3/3-2x^2]_0^4=-32/3$ J.

Worked derivation:

$$W = \int_0^4 (x^2 - 4x)\,dx = \left[\frac{x^3}{3} - 2x^2\right]_0^4 = -\frac{32}{3}\,\text{J}.$$

Why: The force is negative throughout the open interval $0<x<4$, so work must be negative. Integrating gives $[x^3/3-2x^2]_0^4=64/3-32=-32/3$ J. The zeros of force at the endpoints do not cancel the negative area accumulated between them.

Why not the others: (A) uses magnitude but loses sign; (C) assumes endpoint zeros imply zero area; (D) replaces the curved force profile by an incorrect constant.

Key point: The signed area under F(x) determines work.

Q48

Physics · medium

Gravitation / escape speed ratio

Answer: (C) √3v/2.

Given: v²=2GM/R; target radius 4R.

Formula/Principle: ½mu²=GMm(1/R−1/4R).

Step 1: u²=(3/4)v².

Worked derivation:

$$u = \sqrt{\frac{2GM}{R}\left(1 - \frac{1}{4}\right)} = \frac{\sqrt{3}}{2}v.$$

Why: Energy conservation requires initial kinetic energy equal to the increase in gravitational potential from −GMm/R to −GMm/(4R). That difference is 3GMm/(4R). As v²=2GM/R, the required speed is √3v/2, below escape speed because the target is finite.

Why not the others: (A) is local escape speed at 4R, a different quantity; (B) assumes inverse distance scaling of launch speed; (D) omits the square root.

Key point: Separate local escape speed from finite-height launch energy.

Q49

Physics · medium

Mechanical Properties of Solids / Poisson-ratio response

Answer: (D) −0.0006.

Given: ν=0.30, longitudinal strain +0.002.

Formula/Principle: εtrans=−ν εlong.

Step 1: −0.30×0.002=−0.0006.

Worked derivation:

$$\varepsilon_{\perp} = -\nu\varepsilon_{\parallel} = -(0.30)(0.002) = -6 \times 10^{-4}.$$

Why: Poisson ratio ν=−εtrans/εlong. Substituting 0.30 and +0.002 gives εtrans=−0.0006: a tensile rod narrows sideways. Both strains are dimensionless; the negative transverse sign records contraction rather than an additional force.

Why not the others: (A) and (B) multiply by ten; (C) misses the opposite sign.

Key point: Poisson ratio contains a minus sign.

Q50

Physics · easy

Thermal Properties of Matter / cooling comparison

Answer: (A) 3:2.

Given: Ambient 20°C, bodies 80°C and 60°C.

Formula/Principle: |dT/dt|=k(T−Ta).

Step 1: 60/40=3/2.

Worked derivation:

$$\frac{|\dot{T}_1|}{|\dot{T}_2|} = \frac{80-20}{60-20} = \frac{3}{2}.$$

Why: Newton cooling at the initial moment is proportional to the temperature excess over ambient: 80−20=60 K and 60−20=40 K. The rate ratio is 60:40=3:2.

Why not the others: (B) uses absolute temperatures incorrectly; (C) ratios the two object temperatures; (D) ignores different excesses.

Key point: Use temperature excess above the common ambient.

Q51

Physics · medium

Kinetic Theory / pressure from molecular impacts

Answer: (B) $Nm\langle c_x^2\rangle/L^3$.

Given: N molecules in cube volume L^3.

Formula/Principle: $P=Nm\langle c_x^2\rangle/V$.

Step 1: $V=L^3$.

Worked derivation:

$$P = \frac{N(mc_x^2/L)}{L^2} = \frac{Nm\langle c_x^2\rangle}{L^3}.$$

Why: One molecule transfers momentum $2mc_x$ at each collision, with collisions separated by $2L/|c_x|$, giving mean force mc_x^2/L. Sum N and divide by wall area L^2 to obtain $Nm\langle c_x^2\rangle/L^3$. A factor 1/3 is needed only when replacing component mean square by total speed mean square.

Why not the others: (A) misses a length; (C) double-counts the isotropy factor for a component already specified; (D) adds another unwarranted factor.

Key point: Derive pressure from momentum flux normal to the wall.

Q52

Physics · medium

Oscillations / phase and energy at given displacement

Answer: (C) 3E/4.

Given: E=½kA² and x=A/2.

Formula/Principle: U/E=x²/A², K=E−U.

Step 1: U=E/4, K=3E/4.

Worked derivation:

$$K = \frac{1}{2}k(A^2 - x^2) = E\left(1 - \frac{x^2}{A^2}\right) = \frac{3}{4}E.$$

Why: The spring potential at displacement x is E(x²/A²). At A/2 it equals E/4; conservation leaves kinetic energy 3E/4. The speed is still nonzero away from the turning points, where kinetic energy would vanish entirely.

Why not the others: (A) gives potential energy; (B) assumes a linear energy split; (D) is kinetic energy only at equilibrium.

Key point: Energy varies as displacement squared.

Q53

Physics · medium

Electrostatic Potential and Capacitance / dielectric with battery disconnected

Answer: (D) $V_0/3, U_0/3$.

Given: Q fixed; $C_{new}=3C_0$.

Formula/Principle: V=Q/C, U=Q²/(2C).

Step 1: $V_{new}=V_0/3$; $U_{new}=U_0/3$.

Worked derivation:

$$\frac{V_{\text{new}}}{V_0} = \frac{C_0}{3C_0} = \frac{1}{3}, \qquad \frac{U_{\text{new}}}{U_0} = \frac{C_0}{3C_0} = \frac{1}{3}.$$

Why: Disconnection fixes free charge Q. Insertion raises capacitance threefold, so V=Q/C falls to one third and U=Q²/(2C) also falls to one third. The force drawing dielectric inward supplies the energy difference.

Why not the others: (A) reverses both changes; (B) incorrectly keeps voltage fixed after disconnection; (C) gives the battery-connected energy trend with inconsistent voltage.

Key point: Identify the fixed quantity before choosing energy formula.

Q54

Physics · medium

Moving Charges and Magnetism / Magnetism and Matter / force with vector field components

Answer: (A) vB(ŷ−ẑ).

Given: v=v x̂; B=B(ŷ+ẑ); q>0.

Formula/Principle: F=q(E+v×B)=0.

Step 1: v×B=vB(ẑ−ŷ).

Worked derivation:

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} = vB(\hat{\mathbf{y}} - \hat{\mathbf{z}}).$$

Why: The magnetic part per unit charge is v×B=vB(ẑ−ŷ). Electric and magnetic forces must cancel, so E=−v×B=vB(ŷ−ẑ). This is a force-balance condition, not a magnetic-force-only direction question.

Why not the others: (B) adds to the magnetic force; (C) points partly parallel to B instead of cancelling v×B; (D) leaves magnetic deflection.

Key point: Set E+v×B=0 and resolve both components.

Q55

Physics · hard

Electric Charges and Fields / Electrostatic Potential / flux of multi-charge surface

Answer: (B) 4q/ε₀.

Given: Enclosed +2q, −q, +3q; outside −5q.

Formula/Principle: Φ=Qenclosed/ε₀.

Step 1: Qenclosed=4q.

Worked derivation:

$$\Phi_E = \frac{2q - q + 3q}{\varepsilon_0} = \frac{4q}{\varepsilon_0}.$$

Why: Gauss’s law uses only enclosed net charge: 2q−q+3q=4q. The external charge can alter local field distribution on the surface but its total net flux contribution is zero.

Why not the others: (A) misadds charges; (C) includes outside charge in net; (D) reports only the external source.

Key point: Flux through a closed surface depends on enclosed charge.

Q56

Physics · medium

Current Electricity / resistor network node reduction

Answer: (C) 5 Ω.

Given: Two 6 Ω branches A–B; 2 Ω B–C.

Formula/Principle: RAB=(1/6+1/6)$^{-1}$; RAC=RAB+2.

Step 1: 3+2=5 Ω.

Worked derivation:

$$R_{AB} = \left(\frac{1}{6} + \frac{1}{6}\right)^{-1} = 3\,\Omega, \quad R_{AC} = 3 + 2 = 5\,\Omega.$$

Why: The equal parallel pair combines as (6×6)/(6+6)=3 Ω. Current then passes through the 2 Ω series resistor, giving 5 Ω overall. The common A–B endpoints establish parallel connection; the single B–C arm carries their combined current.

Why not the others: (A) omits series arm; (B) treats parallel pair as 2 Ω; (D) adds only one 6 Ω branch to 2 Ω.

Key point: Reduce nodes and series/parallel stages.

Q57

Physics · easy

Ray Optics and Optical Instruments / lens-mirror combination

Answer: (D) 50/3 cm left of lens.

Given: First lens object distance 30 cm, f=10 cm; mirror x=20 cm.

Formula/Principle: First image x=15; mirror image x=25; second pass lens object distance 25.

Step 1: Second pass 1/v=1/10−1/25=3/50.

Worked derivation:

$$v_1 = 15\,\text{cm}, \quad x_{\text{mirror image}} = 25\,\text{cm}, \quad v_2 = \left(\frac{1}{10} - \frac{1}{25}\right)^{-1} = \frac{50}{3}\,\text{cm left}.$$

Why: On the first pass the lens makes an image 15 cm to its right. The mirror lies 5 cm beyond it and forms a virtual image at x=25 cm behind the mirror. Returning rays approach the lens as if from an object 25 cm to its right; the lens then makes a real image 50/3 cm to its left.

Why not the others: (A) is mirror image distance from mirror, not final location; (B) is focal length; (C) stops at the first lens image.

Key point: Track the intermediate image through both optical elements and the return pass.

Q58

Physics · medium

Dual Nature of Radiation and Matter / photocurrent saturation

Answer: (A) Current doubles; stopping potential unchanged.

Given: Frequency fixed, intensity doubled.

Formula/Principle: Isat∝photon flux; eVstop=hf−φ.

Step 1: Isat doubles, Vstop unchanged.

Worked derivation:

$$I_{\text{new}} = 2I_{\text{sat}}, \qquad eV_{\text{new}} = hf - \phi = eV_{\text{stop}}.$$

Why: Twice the photon flux ejects approximately twice as many photoelectrons per time, raising saturated current. Maximum electron kinetic energy depends on photon frequency, unchanged here, so stopping potential stays fixed.

Why not the others: (B) and (C) incorrectly tie stopping potential to intensity; (D) overlooks the larger electron count.

Key point: Intensity controls count; frequency controls maximum energy.

Q59

Physics · medium

Atoms / emission selection

Answer: (C) 4→3.

Given: $E_n = -13.6/n^2$ eV.

Formula/Principle: $\Delta E = 13.6(1/n_{\text{low}}^2 - 1/n_{\text{high}}^2)$, $\lambda = hc/\Delta E$.

Step 1: 4→3 has smallest ΔE.

Worked derivation:

$$\Delta E_{4\to3} = 13.6\left(\frac{1}{9} - \frac{1}{16}\right) < \Delta E_{3\to2}, \Delta E_{4\to2}, \Delta E_{2\to1}.$$

Why: Longest wavelength means smallest emitted energy. Hydrogen gaps shrink among high neighbouring levels: 4→3 has 13.6(1/9−1/16) eV, less than the other displayed drops.

Why not the others: (A) spans two levels and is larger; (B) has a larger adjacent gap lower in the ladder; (D) has the largest of these gaps.

Key point: Compare energy gaps, then invert for wavelength.

Q60

Physics · hard

Electromagnetic Induction / mutual induction sign

Answer: (D) Into the right dot.

Given: Same dot orientation; left-dot entering current decreases.

Formula/Principle: Lenz law: induced right current restores declining linked flux.

Step 1: It enters the right dot.

Worked derivation:

$$\mathcal{E}_2 = -M\frac{dI_1}{dt} > 0 \quad \text{in the chosen dot-entering reference direction.}$$

Why: By the dot convention, currents entering corresponding dotted ends produce fluxes of the same reference sign. A decreasing left-dot entering current reduces positive flux in the right coil, and Lenz's law drives a current entering its dotted end to oppose that reduction.

Why not the others: (A) would reinforce the flux decrease; (B) neglects magnetic coupling; (C) ignores the specified monotonic change.

Key point: Choose a reference winding direction and oppose flux change.